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MAJIBILL PLATFORM · DELIVERY

Delivery Plan, Team & Commercial Strategy

How a startup team of more than four engineers builds and ships MajiBill with agentic workflows: roles, roadmap, quality gates, risks, IP and pricing strategy.

```
spec -> plan -> agent -> tests -> review -> ship
# humans own money, tenancy and migrations
# register the company before you sign
```

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MB-09

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For review

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Founders, tech lead, delivery lead (internal)

CLASS
Confidential

Working codename: MajiBill (Swahili: maji = water). Figures marked 'assumption' must be validated before commitment.

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// 1. Team and roles

The team has more than four capable engineers. The plan assumes **6–8 people**, with agents multiplying implementation speed but not review capacity: the bottleneck is human review of money-critical code, so roles are set up to protect it.

Role	Count	Owns
Tech lead / architect	1	Architecture, ADRs, money-path review, cross-team contracts, client technical liaison
Backend engineers (TypeScript)	2	Core modules: tenancy, billing, ledger, payments, audit; migrations
Platform / DevOps + Go engineer	1	IaC, CI/CD, observability, payment ingress, cells, backups
Frontend engineers (Next.js)	1–2	Staff app, customer portal, platform console, design system
Mobile engineer (Flutter)	1	Offline reader app and sync client
QA / test automation (part-time or shared)	0.5–1	Golden bill tests, E2E, sync harness, load tests
Product / delivery lead (founder or BA)	1	Requirements, client demos, acceptance, tariff and process knowledge

RISK **Single points of failure**
Ledger/billing knowledge and payment integration must each be understood by at least two people. Pair on these modules from day one.

// 2. Agentic workflow (baseline)

The team uses the **baseline** planning-first workflow: durable project context and specs live in the repository, agents read them in a defined order, and humans review at gates. MajiBill adopts it unchanged in spirit and adds money-path rules.

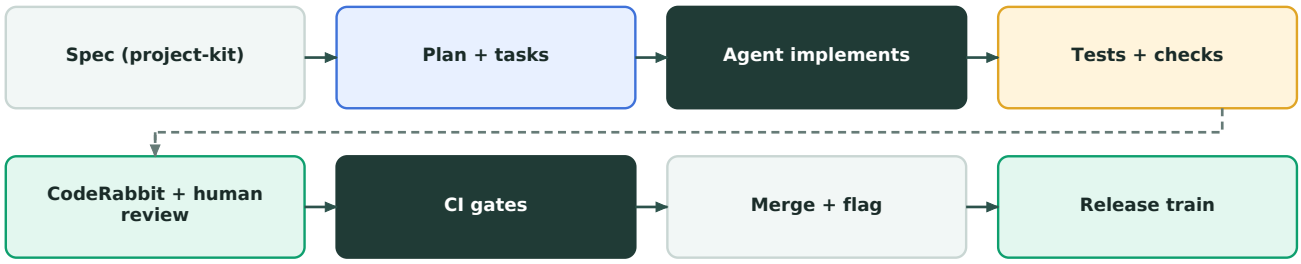


Fig 2.1 Spec-first loop. Specs and decisions are the source of truth; code follows.

Rule	Detail
Spec before code	Every feature has a spec with acceptance criteria, data impact, audit events, permissions and test list (templates in `templates/`)
`AGENTS.md` is law	Reading order, conventions, forbidden actions; kept short and current; updated when an agent makes the same mistake twice
Small tasks	Agents work on tasks sized for a single reviewable PR; no sweeping refactors without an ADR
Protected paths	`modules/ledger`, `billing`, `tenancy`, `identity`, `migrations/`, `payment-ingress`: two human reviewers + golden/property tests; agents may propose, never self-merge
Tests as the contract	Agent must add or update tests; CI rejects coverage drops on domain code; billing changes require updated golden files signed off by a finance-literate reviewer
No dependency drift	Agents may not add dependencies without a note in the PR; supply-chain scan runs on every PR
Bugs captured	Reproductions in `bugs/` with sanitised evidence; each production bug yields a test and, if systemic, an `AGENTS.md` rule
Knowledge base	Shared research (Daraja, tariffs, WASREB) lives in `research/` so agents and humans use one truth

// 3. Roadmap

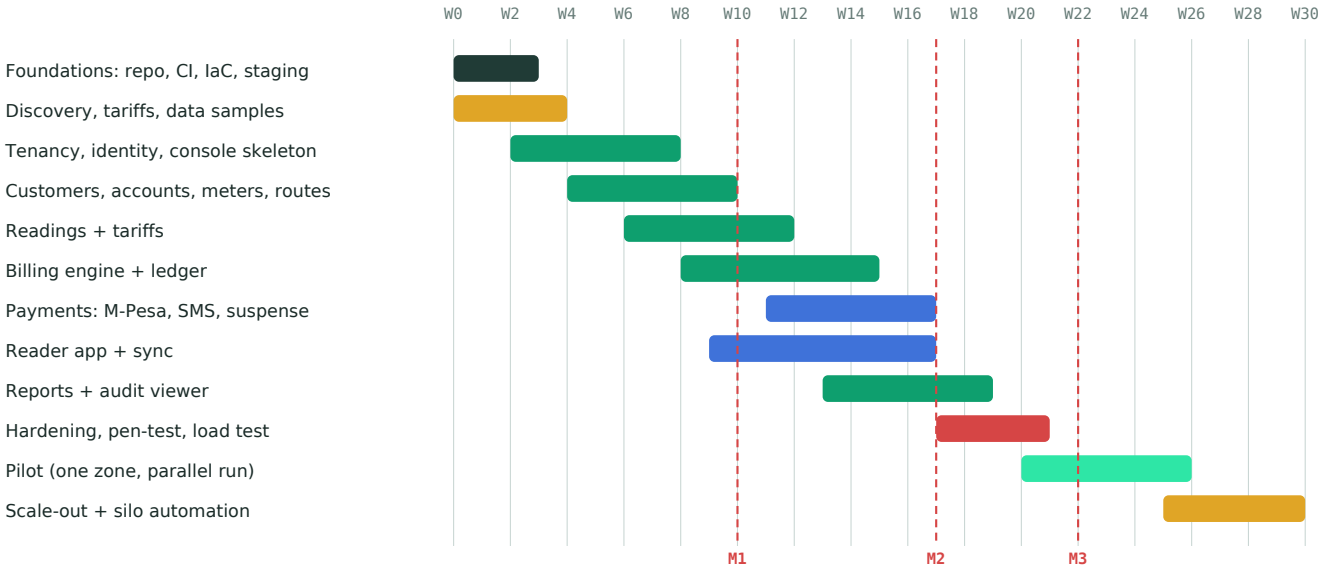


Fig 3.1 30-week plan. M1 demo (W10), M2 feature complete (W17), M3 pilot go-live (W22). Overlaps are intentional.

Phase	Weeks	Exit criteria
P0 Foundations and discovery	0–4	Signed scope; sample data and tariffs received; pipeline green; staging live; contracts and entity ready
P1 Core build	3–15	Demo (M1, W10): customer → reading → bill → manual payment; golden bill tests approved by client finance
P2 Payments and field	9–17	Feature complete (M2): M-Pesa auto-post in sandbox, offline reader app, SMS, suspense workflow, reports
P3 Hardening and pilot	17–26	Pen-test fixed; load test passed; restore drill passed; pilot zone billed in parallel with legacy for 2 cycles; go-live at M3
P4 Scale-out	25+	Second tenant onboarded in pooled cell; silo cell automation; DR drill; cost review

The earlier discussion estimated 5–7 months with 3–5 developers. With 6–8 engineers and agent-assisted delivery, 22 weeks to pilot go-live is plausible **if** scope stays at the MVP in MB-08, client data arrives on time, and Daraja production approval is not on the critical path. Treat it as a target to be validated after P0.

// 4. Quality strategy

Layer	Approach	Gate
Unit / domain	Pure billing and ledger logic, property tests, mutation testing on billing	Block merge below threshold
Integration	Testcontainers with real Postgres; RLS and cross-tenant tests; migration tests on seeded data	Block merge
Contract	OpenAPI diff, generated clients compile (TS, Dart, Go, Python)	Block merge
E2E	Playwright journeys: reading to bill, M-Pesa to receipt, adjustment approval	Block release
Sync / field	Simulated devices: duplicates, reordering, network drops; real-device test on low-end Android	Block release
Performance	k6: 750k-bill run, 100 tx/s payment burst, concurrent reports	Before pilot, then each quarter
Security	Dependency scan, SAST, pen-test, RLS policy-coverage check	Block merge / pilot
UAT	Client finance and operations validate bills against their manual calculations	Acceptance for each milestone

// 5. Risk register

#	Risk	L	I	Mitigation
1	Tariff rules and client data are messier than expected	H	H	Early discovery with real samples; import tooling; tariff model flexible and effective-dated
2	Daraja production access or Paybill setup delays	M	H	Start applications in P0; sandbox-first; per-tenant shortcode runbook; manual payment fallback
3	Ledger or billing bug causes wrong bills	M	H	Golden tests, property tests, parallel run, maker–checker, approval before publish
4	Cross-tenant data leak	L	H	RLS + composite FKs + tests + pen-test; cells for big tenants
5	IP ownership dispute with ZUCAR	M	H	Settle IP and licensing before build; written reseller agreement
6	Company not yet incorporated when contracts are signed	H	M	Register the company, tax PIN and bank account first; no personal-liability contracts
7	Scope creep from “every county is different”	H	M	Config over code; change-request process; product backlog owned by delivery lead
8	Key-person dependency	M	H	Pairing, documentation, ADRs, runbooks; two owners per critical module
9	Agent-introduced defects or insecure code	M	M	Protected paths, human review, CI policy checks, small PRs
10	Field adoption: readers resist the app	M	M	Co-design with readers, offline reliability, training, supervisor dashboards
11	SMS cost overrun	M	M	Per-tenant SMS wallet, caps, templates, segment counting; price pass-through
12	Infrastructure cost exceeds revenue early	M	M	Lean pilot cell, scale on measured load, pooled cells for small tenants

// 6. Commercial and IP strategy (internal)

The question “multitenant SaaS or sell it to the client” is mostly a **contract** question. The architecture supports all models (MB-01); the agreement decides which you are in.

Model	What ZUCAR gets	Who owns IP	Who operates	Revenue shape	Our view
A. Reseller partnership	Right to resell/white-label in agreed territory; first-mover pricing	404HubSpot	404HubSpot	Setup fee + per-connection subscription; partner margin	**Preferred**
B. Licensed dedicated deployment	Dedicated cell or sovereign install for a client	404HubSpot	404HubSpot or client	Licence + setup + annual support/maintenance	Good for large tenants
C. Outright purchase	Source and assignment of IP	ZUCAR	ZUCAR	Large one-off fee; exit from product	Only at a price that replaces the platform’s future value; never exclusive without a premium

> DECISION **Recommendation**
> Offer **A as the base**, **B for tenants that need isolation**, and quote **C** only at a deliberately high price with a field-of-use limit (e.g. ZUCAR’s territory) so you can still serve other markets. Keep the control plane and release channel under your control in all cases.

6.1 Pricing structure

- **One-time setup and onboarding** per tenant: data migration, tariff configuration, training, go-live support.
- **Platform subscription** per active billable connection per month, with tiers (price falls with volume); minimum monthly fee for small tenants.
- **Pass-through** costs: SMS, payment-gateway fees, in-country hosting premiums.
- **Support tiers:** standard (business hours), extended, 24/7 for large tenants.
- **Partner margin** for ZUCAR on subscriptions it sources (a figure to negotiate; a common starting range is 20–30%).
- **Change requests** at a day rate; roadmap items sold as modules (STK portal, analytics, smart meters).

Worked example with **hypothetical** numbers (replace with your cost model): a tenant with 30,000 active connections at KES 20 per connection per month yields KES 600,000 per month before partner margin and pass-throughs; at a 25% partner margin that is KES 450,000 net to 404HubSpot, against infrastructure of roughly 1–2% of that at this size. The leverage is in volume across many tenants, not in any single contract.

6.2 Cost-plus check for the first quote

pricing worksheet

```
price = (dev_effort_pm x blended_monthly_cost
         + infra_first_12_months + security_testing + legal_and_registration
         + discovery_and_training + support_reserve) x (1 + contingency)
         x (1 + target_margin)
# dev_effort_pm: planning assumption 30-40 person-months to pilot (6-8 people, ~5 months)
# contingency: 15-25% for unknown tariffs/data; target_margin: your choice
```

6.3 Contract must-haves

1. IP ownership and licence grant, including field of use and territory for ZUCAR.
2. Data ownership: tenants own their data; processing agreement and sub-processor list.
3. Payment terms tied to milestones (M1, M2, M3) with acceptance criteria from MB-08.
4. Service levels, support hours and credits; limitation of liability.
5. Exit and data export; escrow of source for sovereign licensees if requested.
6. Change-control process; confidentiality; non-solicit; governing law and dispute resolution (Kenya).

> WATCH OUT **Register before you sign**
> Incorporate the company, obtain a tax PIN, open a business bank account, and take counsel’s review of the agreements. Contracts signed informally can leave founders personally exposed and can complicate ODPC registration and Safaricom shortcode applications.

6.4 Governance cadence

Ritual	Frequency	Purpose
Stand-up	Daily, 15 min	Unblock; surface review queue
Sprint planning / review	Fortnightly	Scope, demo to client, acceptance
Architecture review	Weekly, 45 min	ADRs, cross-module changes, protected-path PRs
Risk review	Fortnightly	Update register; escalate
Release train	Weekly after pilot	Controlled releases, changelog, tenant notices
Client steering	Monthly	Milestones, commercial, roadmap